MATH 101 Unit 2: Functions and Their Graphs

Practice 1 | Version 202610122020.37 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

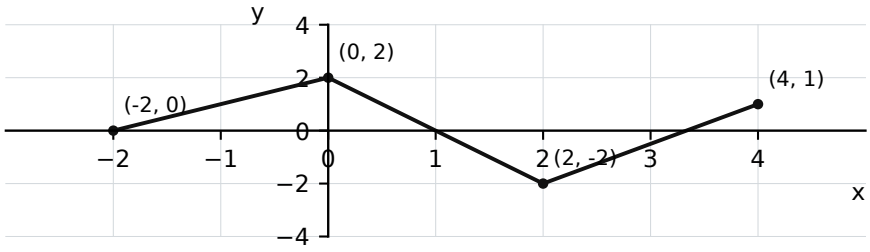
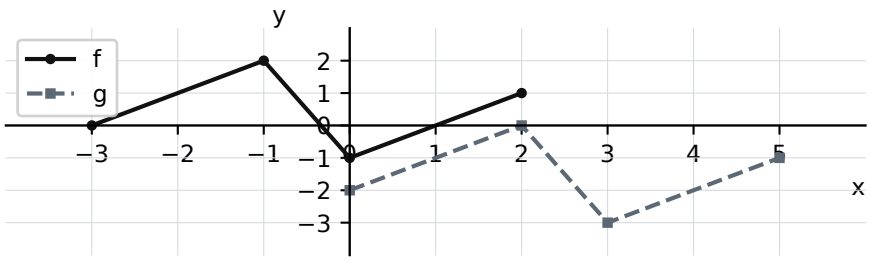
QUESTIONS	ANSWERS
1. (a) Choose the pair giving h(x) = f(g(x)). (b) Form F(G(x)) and evaluate it at the supplied input. $h(x) = (2x + 1)^2 + 3$ $A.\quad f(x) = 2x + 1,\ g(x) = x^2 + 3$ $B.\quad f(x) = 2x^2 + 3,\ g(x) = x + 1$ $C.\quad f(x) = x^2 + 1,\ g(x) = 2x + 3$ $D.\quad f(x) = x^2 + 3,\ g(x) = 2x + 1$ $(b)\quad F(x) = x^2,\ G(x) = x - 2,\ x_0 = 3$	1. (a) Choice 8 pts (b) Composition 8 pts (b) Value 4 pts
2. For this function, find (a) f(-1), (b) domain, and (c) range. $f(x) = 2\lvert x - 1\rvert - 2$	2. (a) Value 6 pts (b) Domain 7 pts (c) Range 7 pts
3. (a) Find the linear function through these points, with domain all real numbers. (b) Find its inverse and give the inverse domain and range. $(1,\ 1);\ (3,\ 5)$	3. (a) Function 10 pts (b) Inverse 6 pts (b) Domain 2 pts (b) Range 2 pts

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4. From the graph, find (a) average rate from x = -2 to x = 4; (b) increasing and decreasing intervals; (c) the two local extrema (ordered pairs). 	4. (a) Rate 6 pts (b) Increasing 4 pts (b) Decreasing 4 pts (c) Local max 3 pts (c) Local min 3 pts
5. The solid graph is f and the dashed graph is g. Select the expression for g(x). *A*. $g(x) = f(x-3) - 2$ *B*. $g(x) = f(x+3) + 2$ *C*. $g(x) = f(x+3) - 2$ *D*. $g(x) = f(x-3) + 2$ 	5. Choice 20 pts

Instructor answer key: MATH 101 Unit 2: Functions and Their Graphs

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a) Choice

D

8 pts

(b) Composition

$x^2 - 4x + 4$

8 pts

(b) Value

1

4 pts

2. (a) Value

2

6 pts

(b) Domain

$(-\infty, \infty)$

7 pts

(c) Range

$[-2, \infty)$

7 pts

3. (a) Function

$f(x) = 2x - 1$

10 pts

(b) Inverse

$f^{-1}(x) = \frac{x}{2} + \frac{1}{2}$

6 pts

(b) Domain

$(-\infty, \infty)$

2 pts

(b) Range

$(-\infty, \infty)$

2 pts

TEST PAGE 2

4. (a) Rate

$\frac{1}{6}$

6 pts

(b) Increasing

$(-2, 0) \cup (2, 4)$

4 pts

(b) Decreasing

$(0, 2)$

4 pts

(c) Local max

$(0,\ 2)$

3 pts

(c) Local min

$(2,\ -2)$

3 pts

5. Choice

A

20 pts